

EXPERIMENT–1

Title

IoE Sensor Data Acquisition and Preprocessing using Python in Google Colab

Aim

To acquire IoE sensor data, perform preprocessing techniques such as handling missing values, removing duplicates, detecting outliers, normalizing data, and preparing the dataset for AI/ML applications using Python in Google Colab.

Objectives

Understand IoE sensor datasets.
Import IoE sensor data into Python.
Perform data cleaning.
Handle missing values.
Remove duplicate records.
Normalize sensor values.
Visualize the processed data.
Prepare the dataset for machine learning.

Software Requirements

Google Colab

Python 3

Pandas

NumPy

Matplotlib

Seaborn

Scikit-learn

Theory

Internet of Everything (IoE)

The Internet of Everything (IoE) extends the Internet of Things (IoT) by integrating four major components:

- People
- Processes
- Data
- Things

Sensor devices continuously generate environmental data such as:

- Temperature
- Humidity
- Pressure
- Gas concentration
- Heart Rate
- Oxygen Saturation
- Motion

IoE Sensor Data Acquisition and Preprocessing

Introduction

Before AI models can use IoE sensor information, the raw sensor data must undergo **data acquisition** and **data preprocessing**. These processes improve data quality and prepare it for Artificial Intelligence (AI) and Machine Learning (ML) applications.

Data Acquisition

Definition:

Data acquisition is the process of collecting information from IoE sensors.

Example Sensors

- Temperature Sensor
- Humidity Sensor
- Motion Sensor
- Gas Sensor
- Heart Rate Sensor
- GPS Sensor

Data Preprocessing

Data preprocessing includes the following steps:

1. Missing Value Handling

Replace missing values using:

- Mean
- Median
- Mode

2. Duplicate Removal

- Remove repeated records.

3. Outlier Detection

Detect abnormal values using:

- IQR (Interquartile Range)
- Z-score

4. Normalization

- Convert values into the same range (0–1).

5. Visualization

Visualize data using:

- Histograms
- Boxplots
- Scatter Plots
- Correlation Matrix

Algorithm

Step 1: Start Google Colab.

Step 2: Import required libraries.

Step 3: Load IoE sensor dataset.

Step 4: Display dataset.

Step 5: Check missing values.

Step 6: Fill missing values.

Step 7: Remove duplicate records.

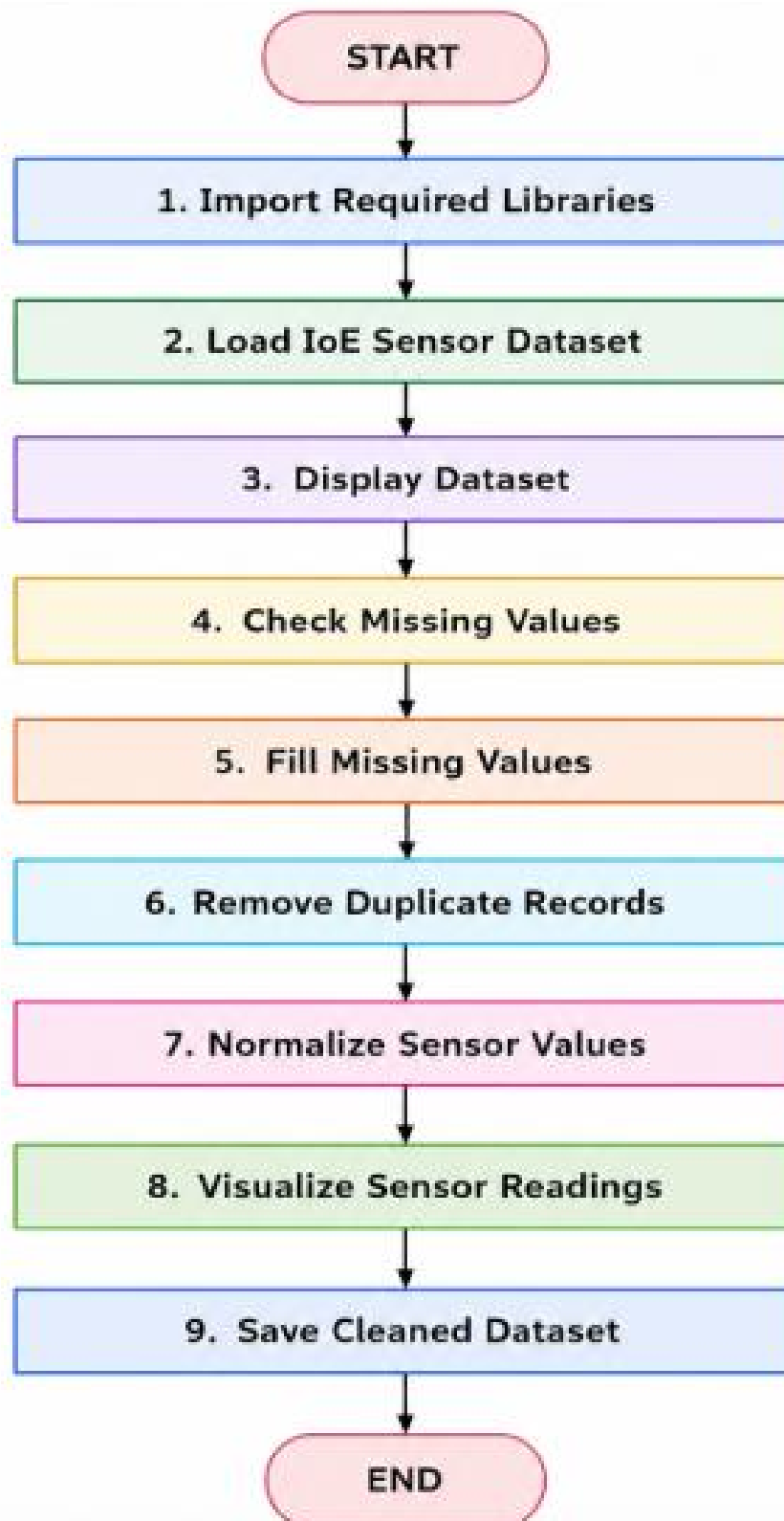
Step 8: Normalize sensor values.

Step 9: Visualize sensor readings.

Step 10: Save cleaned dataset.

Step 11: Stop.

FLOWCHART



PYTHON PROGRAM IN (GOOGLE COLAB)

```
# Import Libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import MinMaxScaler

# Load Dataset

df = pd.read_csv("sensor_data.csv")

# Display Dataset

print(df.head())

# Dataset Information

print(df.info())

# Check Missing Values

print(df.isnull().sum())

# Fill Missing Values

df.fillna(df.mean(numeric_only=True), inplace=True)

# Remove Duplicate Records

df.drop_duplicates(inplace=True)

# Normalize Numeric Columns

scaler = MinMaxScaler()

num_cols = df.select_dtypes(include=np.number).columns

df[num_cols] = scaler.fit_transform(df[num_cols])
```

```

# Display Cleaned Dataset

print(df.head())

# Save Cleaned Dataset

df.to_csv("clean_sensor_data.csv", index=False)

# Plot Temperature Sensor

plt.figure(figsize=(8,5))
plt.plot(df['Temperature'])
plt.title("Temperature Sensor Readings")
plt.xlabel("Samples")
plt.ylabel("Normalized Temperature")
plt.grid(True)
plt.show()

# Correlation Heatmap

plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.show()

```

Sample Dataset (sensor_data.csv)

Temperature	Humidity	Pressure	Heart Rate	SpO ₂
28	65	1012	78	98
29	66	1011	80	99
30	68	1010	81	97
31	69	1013	79	98
29	67	1011	80	99

Expected Output

- ✓ Dataset successfully loaded.
- ✓ Missing values handled.
- ✓ Duplicate records removed.
- ✓ Data normalized.
- ✓ Temperature graph displayed.
- ✓ Correlation heatmap generated.
- ✓ Cleaned dataset saved as `clean_sensor_data.csv`.

Applications

- Smart Healthcare
- Smart Cities
- Industrial IoT
- Smart Agriculture
- Environmental Monitoring
- Smart Home Automation
- Predictive Maintenance

Result

The IoT sensor dataset was successfully acquired, preprocessed, normalized, and visualized using Python in Google Colab. The cleaned dataset is now suitable for Artificial Intelligence (AI) and Machine Learning (ML) applications.

